

VPA IB DP CS — Year 1 Semester 1 Overview

VPA IB DP Computer Science — Year 1 Semester 1 Overview

School: Victoria Park Academy Shenzhen (VPA) | Programme: IB DP Computer Science (First Assessment 2024)

Level: Standard Level (SL) + Higher Level (HL) | Language of Instruction: English with Mandarin scaffolding

Semester Overview

Semester 1 of Year 1 covers Topic 1: System Fundamentals and Topic 2: Computer Organization. These foundational topics establish the conceptual framework for the entire IB DP Computer Science course and build directly on the IGCSE Computer Science (0478) knowledge students have acquired.

Topics Covered

Weeks	Topic	IB Syllabus Reference	Level
1 – 9	System Fundamentals	Topic 1 (1.1 – 1.2)	SL + HL
10 – 18	Computer Organization	Topic 2 (2.1 – 2.5)	SL + HL

Learning Objectives (IB Command Terms)

- By the end of this semester, students will be able to:
- Define key terms related to systems and computer organization (定义 dìngyì)
 - Describe the stages of system development and the components of computer architecture (描述 miáoshù)
 - Explain how data is represented in binary and how the CPU executes instructions (解释 jiěshì)
 - Construct ER diagrams, flowcharts, and logic gate circuits (构建 gòujiàn)
 - Compare different memory types, storage devices, and logic operations (比较 bǐjiào)
 - Evaluate the suitability of different systems for given organizational contexts (评估 pínggū)

Differentiation Strategy

Level	Support	Challenge
Bronze	Bilingual keyword lists, sentence starters, scaffolded note templates	Identify and label diagrams; complete cloze exercises
Silver	Partial scaffolding, guided practice	Explain processes in own words; solve standard problems
Gold	Independent research prompts	Analyse case studies; design original solutions; peer teaching

TOK Connections This Semester

Topic 1: To what extent can we know that a new system will solve the problems it was designed to address? How do models (ER diagrams, flowcharts) represent reality, and what do they omit?

Topic 2: If a computer processes everything as binary, how does meaning emerge from mere 1s and 0s? Is the map (architecture diagram) the territory (the physical machine)?

IA Preparation

During this semester, students will begin IA Stage 0: Contextual Awareness. They will:

Understand what constitutes a real client and a real problem

Begin documenting potential client contacts and problem areas

Review exemplar IAs from the IB

Assessment Schedule

Assessment	Week	Format	Weight (Formative)
Diagnostic Test	1	IGCSE recap + IB readiness	Baseline
Topic 1 Progress Check	5	Short-answer + diagramming	15%
Topic 1 End-of-Topic Test	9	Paper 1-style questions	25%
Topic 2 Progress Check	14	Binary calculations + architecture	15%
Semester 1 Examination	18	Paper 1 mock (1h30 SL / 2h10 HL)	35%
IA Client Proposal	16	Written proposal (Criteria A prep)	10%

Lesson Structure (40 Minutes)

Time	Activity	Purpose
0 - 5 min	Starter / Retrieval practice	Activate prior knowledge
5 - 15 min	Direct instruction with bilingual keywords	Present new concepts
15 - 25 min	Guided practice / Mini-task	Apply immediately
25 - 35 min	Independent or paired work	Consolidate learning
35 - 40 min	Plenary / Exit ticket	Check understanding

Resources

IB DP Computer Science Guide (First Assessment 2024)

Computer Science Illuminated (Dale & Lewis)

Cambridge IB Computer Science Coursebook (Watson & Reyner)

VPA Bilingual Glossary (see Supporting Documents)

IGCSE to IB DP Bridge Document

